

# React Native Complete Roadmap for Android Developers

This document provides a **structured, end-to-end roadmap** for Android developers who want to start and build React Native projects **efficiently and professionally**.

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## Phase 0: Mindset Shift (Android → React Native)

Before writing code, understand the conceptual differences.

| Android                | React Native                     |
|------------------------|----------------------------------|
| Activities / Fragments | Components                       |
| XML layouts            | JSX                              |
| View hierarchy         | Component tree                   |
| Lifecycle methods      | Hooks ( <code>useEffect</code> ) |
| DI (Hilt/Dagger)       | Context / Hooks                  |
| LiveData / Flow        | State / Props                    |
| Gradle                 | Metro + npm                      |

**Core idea:**

UI = Function(State) → JSX

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## Phase 1: JavaScript & ES6 Fundamentals (Mandatory)

You must be comfortable with modern JavaScript.

**Topics to master**

- `let`, `const`
- Arrow functions
- Destructuring
- Spread operator
- Array methods ( `map`, `filter`, `reduce` )
- Promises and `async/await`
- Modules ( `import` / `export` )

**Android mapping** | Android | React Native | |-----|-----| | Kotlin data class | JavaScript object | |  
Extension functions | Utility functions | | Coroutines | async/await |

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## Phase 2: React Fundamentals

These concepts form the foundation of all React Native apps.

- JSX syntax
  - Functional components
  - Props
  - State ( `useState` )
  - Event handling
  - Conditional rendering
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## Phase 3: Core React Hooks (Critical)

Hooks replace most lifecycle and architecture patterns you know from Android.

### Essential hooks

- `useState`
- `useEffect`
- `useRef`
- `useMemo`
- `useCallback`
- `useContext`

**Android mapping** | Android | React | |-----|-----| | `onCreate` | `useEffect` | | `ViewModel` | Custom hook | |  
`LiveData` | `State` |

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## Phase 4: React Native Core Components & Styling

### Core components

- `View`
- `Text`
- `Image`
- `ScrollView`
- `FlatList`
- `Pressable`
- `SafeAreaView`

## Styling topics

- Flexbox (row / column)
  - Padding & margin
  - Borders & radius
  - Shadows (Android vs iOS)
  - Image styling & backgrounds
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## Phase 5: Navigation (Activities Equivalent)

### Library

- React Navigation

### Concepts

- Stack navigation
- Bottom tab navigation
- Drawer navigation
- Passing parameters
- Nested navigation

**Android mapping** | Android | React Native | |-----|-----| | Activity stack | Navigation stack | | Intent extras | Route params |

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## Phase 6: State Management

### Start with

- Local state ( `useState` )
- Lifting state up
- Context API

### Advanced (when required)

- Redux Toolkit
- Zustand
- Recoil

**Android analogy** | Android | React Native | |-----|-----| | ViewModel | Store | | Repository | Service layer |

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## Phase 7: Networking & API Integration

- `fetch`
  - Axios
  - REST APIs
  - Headers & authentication tokens
  - Error & loading handling
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## Phase 8: Forms & Input Handling

- `TextInput`
- Keyboard handling
- Formik
- Yup validation

**Android mapping** | Android | React Native | |-----|-----| | `EditText` | `TextInput` | | `InputFilter` | Validation schema |

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## Phase 9: Device & Native Features

- Camera
- Permissions
- Location
- File system
- Push notifications

Common libraries: - `react-native-permissions` - `react-native-image-picker` - `react-native-firebase`

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## Phase 10: Android-Specific Knowledge (Your Advantage)

Leverage your Android background.

- Android folder structure in RN
  - `MainActivity` and `MainApplication`
  - `AndroidManifest.xml`
  - Gradle configuration
  - Native module linking
  - Build variants & flavors
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## Phase 11: Performance Optimization

- FlatList optimization
  - `keyExtractor`
  - Memoization
  - Avoid unnecessary re-renders
  - Image performance tuning
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## Phase 12: Project Architecture (Production Ready)

### Recommended structure

```
src/  
  components/  
  screens/  
  navigation/  
  services/  
  hooks/  
  utils/  
  assets/
```

### Principles

- Separation of concerns
  - Reusable components
  - Feature-based organization
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## Phase 13: Testing

- Jest
  - React Native Testing Library
  - Snapshot testing
  - API mocking
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## Phase 14: Build & Release (Android Focus)

- App signing
- APK vs AAB
- Environment configuration
- CI/CD (GitHub Actions)
- Play Store release process

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## Phase 15: Real-World Projects (Do Not Skip)

Build projects in increasing complexity: 1. Login + API-based app 2. List-detail application 3. Image upload app 4. Offline-first app 5. Production-scale application

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## Suggested Learning Timeline

| Week | Focus                     |
|------|---------------------------|
| 1    | JavaScript + React basics |
| 2    | UI & styling              |
| 3    | Navigation & APIs         |
| 4    | State management & forms  |
| 5    | Native features           |
| 6    | Performance & build       |

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## Final Advice

Do not treat React Native as Android with JavaScript. Master React fundamentals first; mobile expertise follows naturally.

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**This roadmap is sufficient to take an Android developer from zero to production-ready React Native applications.**